

Problem 2.4.1 Robotic Symphony

PLTW

ROBOTIC SYMPHONY



Putting It All Together



Unit 2 Reflection

Putting It All Together

GOALS	MATERIALS	RESOURCES
• Apply knowledge of robotics to a musical performance.		
GOALS	MATERIALS	RESOURCES
• VEX® V5 POE Kit • Consumable resources or recyclables		
GOALS	MATERIALS	RESOURCES

- [Decision Matrix Template](#)
- [Problem 2.4.1 Robotic Symphony Rubric](#)
- [Citing Sources](#)
- [Technical Writing](#)
- [Python® Reference](#)
- [Problem 2.4.1 Robotic Symphony \(Downloadable PDF\)](#)

You and your classmates are tasked with creating a robotic system that will perform a piece (or part of a piece) of music. You will divide into teams, and each team will use the engineering design process to develop a robotic musical instrument. You may use VEX® parts, as well as everyday household items or consumables.

Deliverables and Project Portfolio

Refer to the [Problem 2.4.1 Robotic Symphony Rubric](#) for success criteria for each of these deliverables.

1

Problem Definition. Fully articulate the problem definition, including your decisions, scope, and success criteria.

- As a whole class, you need to decide what piece of music you are going to perform and the instruments that make up your orchestra. Think of songs that will be feasible to reproduce and easy for your

audience to recognize, and the sounds that make up those songs. Using your robotics kit and consumables, you can construct robots to simulate existing musical instruments or create new musical instruments.

- Decide on a plan for how your instrument team will coordinate with the other instrument teams to perform music together. Your teacher will help facilitate this group discussion. This includes but is not limited to:
 - Starting and stopping at the same time
 - Playing in unison
 - Matching volumes and tones in the key of C Major
- Define your team's problem scope, success criteria, and the specific role each member will play on your team. Clearly define what sounds your team is responsible for producing, how you will start and stop producing sound with the rest of the orchestra, and what portions of the selected song your instrument will play. Highlight ways that each individual team member will contribute to the completion of the defined scope of work.



Musical Instruments

Musical instruments fall into four major categories and use different mechanisms to produce sound.

- *Aerophones* use vibrating air.
- *Chordophones* have one or more strings stretched between fixed points.
- *Idiophones* are struck but do not have strings or membranes.
- *Membranophones* have one or more membranes that vibrate.

2

Research. Collect information about a variety of musical instruments. Document the instruments that are of interest to your team and cite all resources in your engineering notebook.

- Image of the instrument
- Name of the instrument
- How it produces sound (aerophone, chordophone, idiophone, or membranophone)
- Country of origin
- Other information you find interesting

NOTE: To review how to properly cite sources in your engineering notebook, visit [Citing Sources](#).

3

Brainstorm. Brainstorm at least three potential robotic solutions to produce the sounds defined in your team's problem scope. You may design a robot that plays an actual instrument or produces sound using household consumables. Create a sketch for each potential solution.

4

Decision Matrix. Use a [Decision Matrix Template](#) to select the best solution. Share your proposed solution with other instrument groups.

5

Technical Drawing. Create a technical drawing that effectively models a mechanism from your solution.

6

Prototype 1. Construct a functioning prototype. Use the [Python® Reference](#) resource if you encounter any issues while writing your code.

7

Test and Evaluate. Design a method to test the [efficacy](#) of your prototype. Conduct the test. Evaluate whether or not your prototype meets the success criteria you outlined in step 1. Reflect on areas that need adjustment.

8

Revise Design. Use the results of your test to modify your robotic construction and code. Modify your existing prototype or construct a new prototype based on your test results. Perform more tests and iterations to improve your robot as needed.

9

Prototype 2. Construct a final version of your robot. This robot should meet all or nearly all the success criteria that you outlined in step 1.

Performance and Solution Documentation. As a whole class, perform your chosen piece of music. Document your final solution.

Unit 2 Reflection

Use the following prompts to write a reflection of your experience in the unit problem. Use the [Technical Writing](#) resource to write your narrative.

Question 1

Describe the robotic orchestra, your team's contribution, and your specific contribution to this end-of-unit problem. What parts of the solution were you most proud of? What parts do you think you would have changed were you to repeat this unit problem?

Question 2

To what extent did you iterate on your design throughout this design exercise? If given another opportunity to iterate, what might you do differently and why?

Question 3

In what parts of this design exercise were you required to make decisions that resulted in a trade-off? How do you think practicing engineers balance competing needs in real-world situations? Explain your reasoning.

Question 4

Reflect on the teamwork aspects of your performance throughout the unit problem. How was your collaboration experience different in this unit problem than in other end-of-unit problems? Specifically, call out areas where you were successful and areas you could improve on next time.